

TANVIR HOSSAIN

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SUMMARY

Hardware security engineer (Ph.D. candidate, expected 2026) with 4+ years of hands-on fault injection and power/EM side-channel analysis against cryptographic and embedded targets. Finds and helps close chip-level security gaps through design review, hardware reverse engineering, and lab-based security testing, and builds ML-guided fault-injection tooling that scales vulnerability discovery across devices. Pairs low-level C, LLVM, and RTL/FPGA fluency with a published record of hardware attacks and mitigations; Champion of the IEEE HOST 2023 Microelectronics Security Challenge.

Core: Fault Injection ♦ Power/EM Side-Channel Analysis ♦ Hardware Reverse Engineering ♦ Threat Modeling ♦ Hardware Trojans ♦ Applied Cryptography ♦ Secure Coding (C, LLVM) ♦ RTL/FPGA ♦ Python, Bash

TECHNICAL SKILLS

Fault Injection & Side-Channel	Power/EM side-channel analysis and countermeasure evaluation for cryptographic implementations; fault-injection attack and mitigation evaluation; ML-guided fault-injection parameter optimization; ChipWhisperer Husky; Keysight/Riscure Spyder, Inspector, Pinata; high-precision EM probes; PicoScope; Tektronix DPO70404C; Analog Discovery 2.
Reverse Engineering & Attacks	Identification and extraction of security primitives from gate-level netlists; hardware Trojan design, insertion, and detection; authentication bypass of PUFs and unlocking of obfuscated scan chains; zero-trust threat modeling of design-for-security primitives; IP protection.
Cryptography & Secure Coding	Software-based protection of cryptographic and arithmetic workloads via residue number coding (C library and LLVM-based program transformations); side-channel and fault countermeasures for lightweight cryptography (ASCON); golden-chip-free, ML-assisted Trojan detection and localization.
Design, EDA & Embedded	RTL design and verification (Verilog, VHDL); Cadence Genus, Encounter, Virtuoso; Xilinx Vivado, ISE; Intel Quartus; ModelSim; STM32, ESP32, Arduino, Raspberry Pi; Xilinx Artix-7, Intel MAX10, and Altera DE2-115 FPGA boards.
Languages & Tools	C (low-level/embedded), Python, Bash, Tcl, MATLAB, Assembly (MIPS); LLVM compiler-pass development; gem5, Intel VTune, McPAT; Git.

EXPERIENCE

Keysight Technologies

May 2026 – Dec 2026

Machine Learning Security Intern, Applied ML & Fault Injection

Santa Rosa, CA

- **Scalable fault-injection tooling:** built model-guided optimization and adaptive-learning workflows that automatically tune fault-injection parameters, improving vulnerability-detection coverage over baseline sweep-based flows (GlitchLab, arXiv 2026).
- **Experiment and telemetry design:** designing experiments, telemetry instrumentation, scalable hyperparameter search, and reliability metrics; delivering a prototype demonstration to the security-test team.

The University of Kansas

Jan 2022 – Present

Graduate Research Assistant, Hardware Security (Advisor: Dr. Tamzidul Hoque)

Lawrence, KS

- **Side-channel and fault attacks on cryptography:** demonstrated power side-channel and fault attacks on the ASCON lightweight-cryptography standard and evaluated countermeasures.
- **Pre-deployment security testing:** proposed a gray-box evaluation methodology that uses power and EM side-channel measurements to flag anomalous behavior in opaque hardware under zero-trust supply-chain assumptions.
- **ML-assisted Trojan detection:** developed golden-chip-free, ML-assisted side-channel techniques for detecting and localizing hardware Trojans on FPGAs (Journal of Hardware and Systems Security 2026; IEEE NAECON 2023).
- **Attacks on chip-level defenses:** reverse-engineered and systematically attacked design-for-security primitives under zero-trust threat models; demonstrated authentication bypass of ring-oscillator PUFs and unlocking of dynamically obfuscated scan chains (ACM TODAES 2026).
- **Reverse-engineering tooling:** built an automated framework that identifies and extracts security primitives from gate-level netlists, mapping a design's defensive surface for pre-silicon security review (GLSVLSI 2026).
- **Software-side mitigation:** built HOACS, a C-library and LLVM-based transformation that keeps secrets encoded during execution on untrusted commodity processors, and evaluated the hardware-Trojan attack surface of lightweight homomorphic obfuscation for cryptographic workloads (IACR TCHES 2026).
- **Training and mentoring:** delivered hands-on side-channel security-testing tutorials and workshops; mentored four NSF REU undergraduates across two universities; co-taught a hands-on hardware course using FPGAs and microcontrollers.

EDUCATION

Ph.D. in Electrical Engineering (Hardware Security) , The University of Kansas	Expected 2026
Dissertation research: fault-injection and side-channel security testing; reverse engineering and adversarial evaluation of design-for-security primitives; ML-assisted vulnerability detection.	
M.S. in Electrical Engineering , The University of Kansas	May 2025
B.Sc. in Electrical and Electronic Engineering , Ahsanullah University of Science and Technology	2016 – 2021

SELECTED PUBLICATIONS

17 peer-reviewed publications (7 journal articles, 10 conference papers) and 1 preprint; full list on [Google Scholar](#).

1. **Tanvir Hossain**, Abhinav Mahadevan, Jasper Van Woudenberg, Rajesh Velegati, and Arindam Bhattacharyya, “GlitchLab: A Hardware-in-the-Loop Optimizer for Physical Fault Injection,” *arXiv preprint*, arXiv:2609.00502 [cs.CR], 2026. [arxiv.org/abs/2609.00502](#)
2. **Tanvir Hossain**, S. M. Mojahidul Ahsan, Prabuddha Chakraborty, Swarup Bhunia, and Tamzidul Hoque, “Attacking the Protections of Hardware using Trojan-Assisted Meta Attacks,” *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2026. doi:10.1145/3804452
3. **Tanvir Hossain**, Matthew Showers, Mahmudul Hasan, and Tamzidul Hoque, “On the Security of Lightweight Homomorphic Obfuscation for Protecting Against Hardware Trojans,” *IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES)*, vol. 2026, no. 2, pp. 736–763, 2026. doi:10.46586/tches.v2026.i2.736-763
4. **Tanvir Hossain**, Ashutosh Ghimire (joint first author), Fathi Amsaad, Tamzidul Hoque, and Ahmed Sherif, “AI-Assisted Side-Channel Analysis for Golden-Chip Free Detection and Localization of Hardware Trojans,” *Journal of Hardware and Systems Security*, 2026. doi:10.1007/s41635-026-00182-4
5. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “SPHINX: A Framework for Security Primitive Hardware Identification and Extraction,” *Great Lakes Symposium on VLSI (GLSVLSI)*, 2026. doi:10.1145/3787109.3815307
6. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “Potential and Pitfalls of Multi-valued Logic Circuits for Hardware Security,” *IEEE Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. doi:10.1109/DCAS57389.2023.10130261
7. Sriram Prabhakara Rao, Shree Ram Abayankar Balaji, Farishta Rahman, Tasneem Suha, Tanzim Mahfuz, **Tanvir Hossain**, Mahmudul Hasan, Md Sakib Hasan, Tamzidul Hoque, Prabuddha Chakraborty, and Prakash Ranganathan, “Secure and Trustworthy Microelectronics: Vulnerabilities, Solutions, and Trends,” *Journal of Hardware and Systems Security*, vol. 9, pp. 163–189, 2025. doi:10.1007/s41635-025-00165-x

SELECTED HONORS & AWARDS

May 2023	Champion, Microelectronics Security Challenge (IP Security) . IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023): first place in the IP-security track of HOST’s hardware security competition.
2022, ’23, ’26	NSF Travel Grants , IEEE HOST Symposium (awarded three times).
Jan 2024	First Place, Poster Presentation , I2S Student Research Symposium, University of Kansas.
Apr 2023	David D. and Mildred H. Robb Award , EECS Department, University of Kansas.
Mar 2021	Best Paper Award , 23rd International Conference on Computer and Information Technology (ICCIT 2020).

TALKS, TRAINING & PROFESSIONAL SERVICE

Invited Talks & Tutorials	<i>Leakage Unplugged: A Hands-On Tutorial on Power Side-Channel Attacks</i> , I2S Student Research Symposium, 2025. <i>Power Side-Channel and Fault Attacks on ASCON and their Countermeasures</i> , Information Advantage Scholars Visit, I2S, University of Kansas, 2024.
Teaching & Outreach	Graduate Teaching Assistant, digital logic and computer architecture (Fall 2024); co-instructor, hands-on computer hardware course using FPGAs and microcontrollers (Fall 2023); NSF-funded hardware summer camps for high-school students (2023, 2024).
Reviewer / TPC	Technical Program Committee, IEEE ISVLSI 2026; reviewer for IEEE Embedded Systems Letters, Journal of Hardware and Systems Security, and the ASEE Annual Conference (2024, 2026).